ASPE Student Challenge 2025

Revision date: Oct 23, 2025

Sponsored by

Some aspects may be modified

1. Overview

This year's Student Challenge involves recording a 30-second song on a 50 mm diameter 3 mm thick acrylic disc that will be played at 33 rpm. For this purpose, a diamond cutting tool, an air-bearing spindle, and a flexure-based X-stage shown in Figure 1 will be utilized. The spindle uses a frameless motor and a rotary encoder to feedback control its rotation at 3.3 rpm (one-tenth of playing speed) or lower. The flexure stage will feed the cutting tool transversely (x) relative to the spindle axis (z) using a voice-coil actuator with its translation in x measured using an optosensor that you can calibrate against a micrometer. A cutting tool holder with its z -axis adjustment to set the depth of cut is attached to top of the X-stage as shown in Figure 2(c). The wave spring should be in its fully compressed state to avoid any play of the tool holder. To self-center the acrylic disc to the spindle axis, the disc is attached to a face plate using a ¼-20 countersunk screw, and then the face plate is attached to the spindle using two 8-32 countersunk screws. Typical music groove spacing on the disc is 125 μm, which corresponds to X-stage travel speed of 6.7 μm/s at 3.3 rpm spindle rotation. In preparation for the final event, your task is to implement a low-velocity position feedback or open loop control the flexure stage shown in Figure 2(a) and (b) for a travel range of ±2.5 mm.

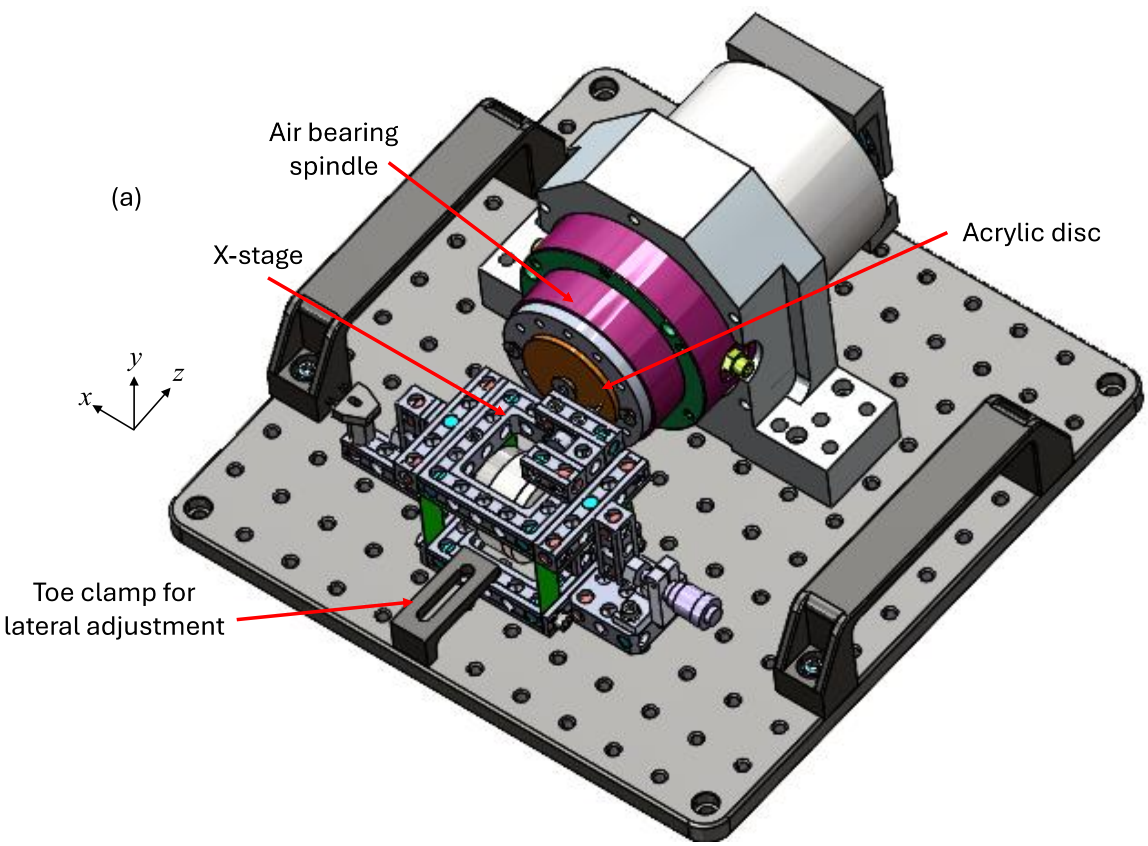


Figure 1: Solid model of the overall setup.

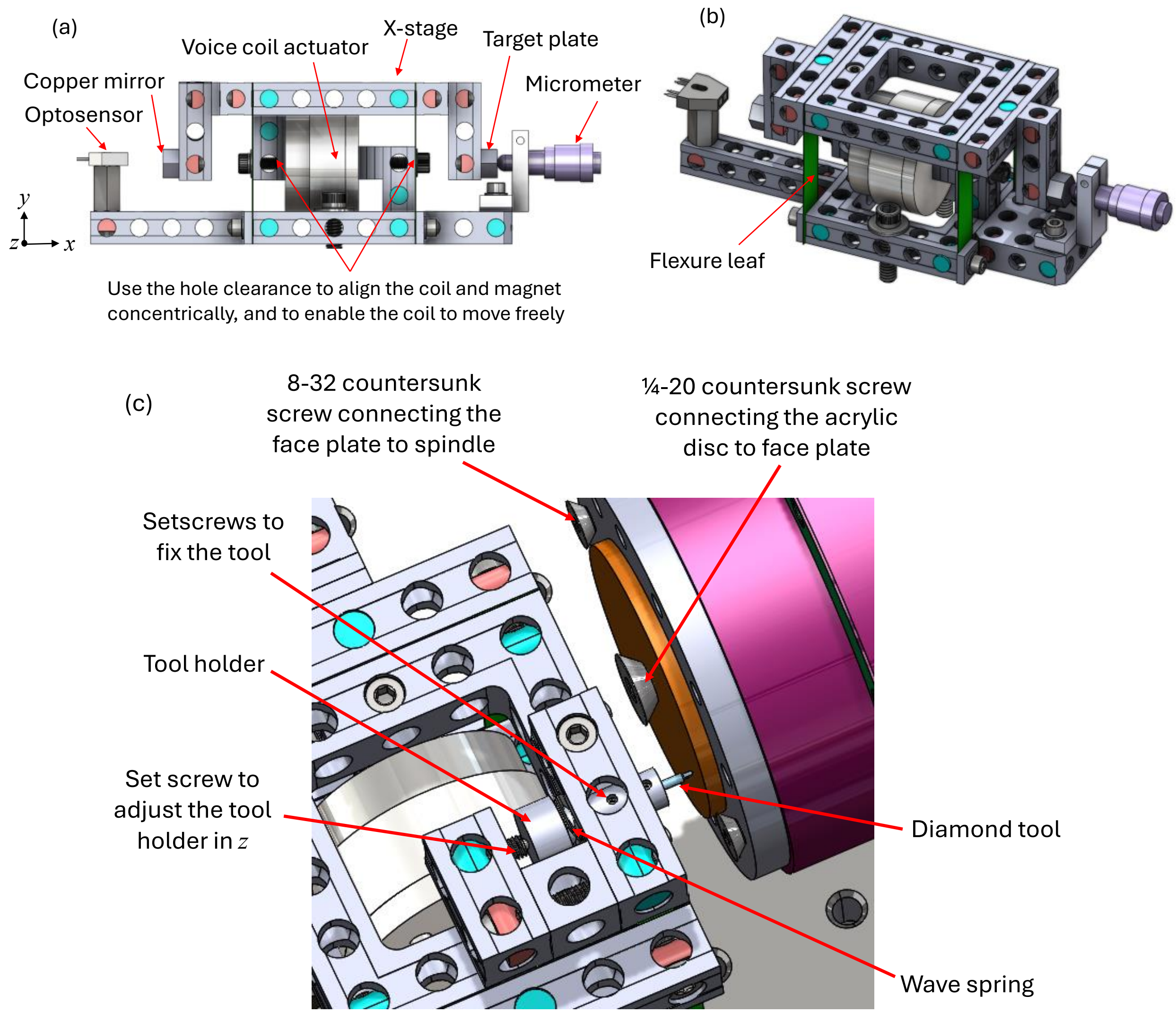


Figure 2: (a, b) Solid model of the flexure-based X stage with its sensor and actuator, (c) adjusting mechanism to set the depth of cut in z axis.

2. Committee-provided components

Teams will receive the following X-stage related components (by Sept. 15), also see sections 7 and 8.

1. Leaf flexures and 'mechblocks' from Motus Mechanical, and CAD model for assembly (Github).
2. NI myRIO-1900 FPGA microcontroller (provided based on request).
3. Voice-coil actuator (LA18-12-006Z), optosensor (OPB742WZ), and micrometer (for displacement control and calibration).

4. ‘Control box’ with 16-bit ADCs and DACs, and 1.5 A transconductance amplifiers.
5. LabVIEW project (GitHub) with examples .vi’s including myRIO–control box communication.

Please contact the committee if you wish to use any additional components not provided by us. Use of components without approval will unfortunately result in a penalty.

3. Optosensor calibration and feedback control of the X-stage

Assemble the X-stage components on your own optical breadboard or by any other suitable means. The mechblocks have 0.5 in. hole spacing and can accommodate either ¼-20 or M6 screws. Once assembled, calibrate the optosensor voltage as a function of micrometer displacement for a ±2.5 mm travel by maintaining contact with the ‘target plate’.

- +X direction: rotate micrometer clockwise to push the target plate; record signal.
- –X direction: rotate counterclockwise and use the voice coil actuator to contact the micrometer tip; record signal.
- Apply a curve fit to the recorded signal to account for sensor nonlinearity.
- After calibration, implement feedback control to follow a ±2.5 mm amplitude triangle wave at 6.7 μm/s for one period, see Figure 3 . To generate this signal, a LabVIEW file, Triangle wave signal.vi is provided in the LabVIEW project.

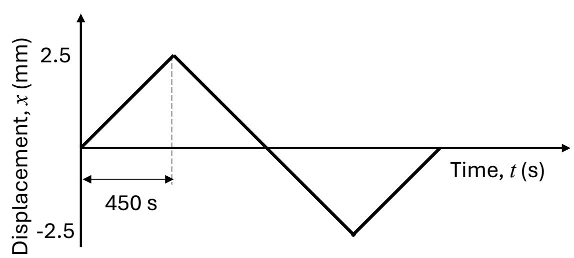


Figure 3: Triangle wave to follow using feedback control

These are all the experiments you are expected to perform prior to bringing your apparatus to the final event at San Diego. A representative X-stage mechanism exhibits a natural frequency of 17 Hz and a damping ratio of 0.09, as determined from a step response using the voice coil actuator. Teams may implement passive damping, if it can be easily removed from the mechblocks before returning them to the committee.

4. Cutting the musical record

At the Final event, teams will integrate the X-stage with the committee-supplied diamond tool, tool holder with its z-adjust mechanism, and spindle to cut the records. The X-stage will ramp from position -2.5 mm to 2.5 mm to produce the music groove spiral as the spindle rotates at 3.3 rpm or lower at the discretion of each team. It will be up to each team to create a control strategy to superimpose the song signal on to the ramp signal by either feedback control or open-loop control. Each team will have its own setup to use until midnight on Monday, Nov 3rd. You will be provided ten acrylic discs, with both faces available for cutting, and each face can accommodate two 30-second songs. Record player(s) will be available for testing. In the event of unexpected equipment failure, or limited availability, teams are expected to share spindles and other essential equipment.

5. Converting music file to drive signal

This section outlines a method for exporting your music as a drive signal to the voice coil actuator of the X-stage. You are welcome to use alternative approaches as needed.

Use websites or any software of your choice to generate a 30 second mono audio file in .wav format, with a sample rate of your choice (e.g., 8 kHz). Convert this .wav file into a .csv file using the **wav to csv.vi** included in the LabVIEW project available on Github. Copy the resulting .csv file to a flash drive and connect it to the myRIO. The contents of the flash drive can be seen via myRIO's default IP address: http://172.22.11.2/files (Username: admin; leave the password field blank). The **read USB and sing.vi** demonstrates how to use this .csv file as an input data for the DAC that drives the voice coil amplifier. This should enable you to play your song using the flexure stage as a speaker. You can integrate this into the program where you've implemented your control strategy to cut the record.

You may also apply the inverse of the frequency response function of the flexure stage, along with the RIAA (Recording Industry Association of America) recording equalization to the .csv file data. An example Excel file, **WW_8kHz_Ztrnsf.xlsx,** which incorporates the models discussed in the following subsections, has been uploaded to Github.

5.1 Inverse frequency response function

The inverse frequency response function of the flexure stage is given by

$$m\ddot{x} + b\dot{x} + kx = u,$$

where x is the raw input signal, u is the transformed output signal. m, b, and k are equivalent mass, damping coefficient, and stiffness respectively. It can approximated using Z-transform as

$$\frac{u}{x} = m\left(\frac{1-z^{-1}}{T}\right)^2 + b\left(\frac{1-z^{-1}}{T}\right) + k,$$

where T is the sampling time, e.g., 125 μs for a song sampled at 8 kHz.

Expanding the above equation,

$$\frac{u}{x} = \frac{m}{T^2}(1-2z^{-1}+z^{-2}) + \frac{b}{T}(1-z^{-1}) + k$$

Each datapoint of the output signal can be written as

$$u_i = \left(\frac{m}{T^2}+k\right)x_i + \left(-2\frac{m}{T^2}+\frac{b}{T}\right)x_{i-1} + \frac{m}{T^2}x_{i-2}.$$

5.2 RIAA equalization

When a music signal is directly used to cut the record, low frequency component (bass) will result in larger wiggles of the groove. To better utilize space occupied by the grooves on the record, during commercial record manufacturing $(\mathcal{R})$, RIAA 'equalization' shown in blue in Figure 4 is used. This suppresses bass and amplifies higher frequencies (treble) to achieve a better signal to noise. When playing $(\mathcal{P})$ the record using the record player, a built-in filter to reverse this equalization is utilized, as shown by the red curve in Figure 4.

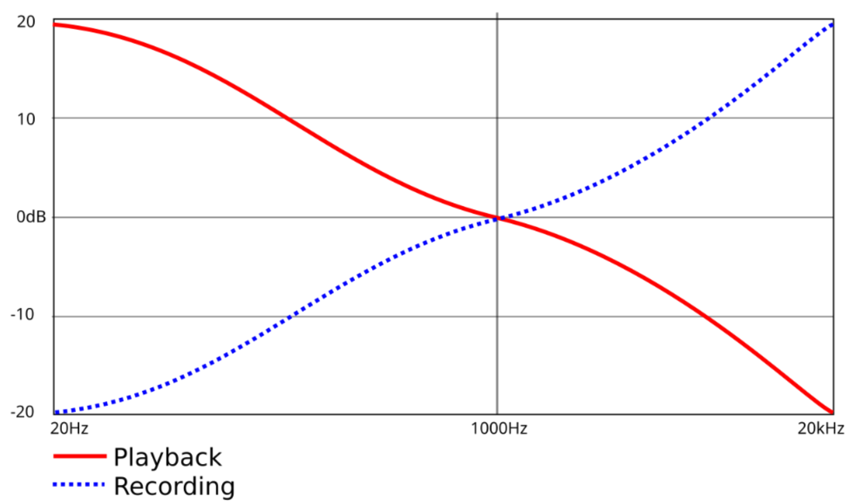


Figure 4: RIAA equalization curve

The playback equalization function is given by

$$\frac{\mathcal{P}}{\mathcal{R}} = \frac{(sT_1+1)(sT_3+1)}{sT_2+1},$$

where T_1 = 3180 μs, T_2 = 318 μs, and T_3 = 75 μs correspond to 2122 Hz, 500.5 Hz and 50.05 Hz.

The converse of this response function called the recording equalization applied during record manufacturing is given by

$$\frac{\mathcal{R}}{u} = \frac{sT_2 + 1}{(sT_1 + 1)(sT_3 + 1)}$$

Using Z-transform, this becomes

$$\frac{\mathcal{R}}{u} = \frac{(1 - z^{-1})T_2 + T}{\big((1 - z^{-1})T_1 + T\big)\big((1 - z^{-1})T_3 + T\big)}$$

Expanding

$$\frac{\mathcal{R}}{u} = \frac{(T + T_2) - T_2 z^{-1}}{T_3 T_1 z^{-2} - (2T_1 T_3 + T_3 T + T T_1) z^{-1} + (T_1 T_3 + T_1 T + T_3 T + T^2)}$$

Each datapoint of the recording signal can be written as

$$R_i = \frac{(T + T_2)u_i - T_2 u_{i-1} - T_3 T_1 R_{i-2} + (2T_1 T_3 + T_3 T + T T_1) R_{i-1}}{(T_1 T_3 + T_1 T + T_3 T + T^2)}$$

6. Total harmonic distortion

In addition to cutting a 30-second song of your choice, you should also produce a record (or records) containing three sinusoidal signals at 250 Hz, 500 Hz, and 2000 Hz, each lasting 5 seconds. The song will be evaluated qualitatively, while the sinusoidal tracks will be analyzed for total harmonic distortion using a microphone and a commercial measurement device.

7. Milestones

- **Preliminary presentation on Zoom (mid-Oct.):** Grading will be based on progress report, effort, and a preliminary error budget. Committee feedback will be provided to enhance the performance.

- **Project report (Friday Oct. 31 midnight):** Submit findings, implementation details, and final error budget in less than 2 pages.

- **Final event (Monday Nov. 3, 6 pm)**: The Student Challenge room (Gallery) would typically be open for the participants to set up their apparatus by noon.

- **Rapid-fire presentations (Wednesday Nov. 5, lunch):** Present results, error budget, and play your record. Each team will have 5 minutes total, including audience questions.

6. Resources

GitHub

https://aspe.net/2025-student-challenge/

https://mail.google.com/chat/u/1/#chat/space/AAQAXU64Upk

aspe.challengecommittee@gmail.com

7. Mecahnical hardware provided

Item	CAD file	Qty.	Comments
Vocie coil BEI Kimco **LA18-12-006Z**		1	
Micrometer		1	
myRIO		upon request	
Mech blocks			
5 x 5	B11-5-5-N	2	
5 x 1	B11-5-1-N	3	
3 x 1	B11-3-1-N	3	One extra
2 x 1	B11-2-1-N	2	
3 x 3	B11-3-3-N	1	
1 x 1/8	B10-1-0.25-EC	10	Two extra
1 x 1		1	One extra
Misc			
CP	CP-10-0.5-FL	1	Target plate
Mirror		1	In CAD, CP-10-0.5-FL used as placeholder
CG	C11-1.5-9.5-N	1	Micrometer holder
PS	Power supplies	2	
PS cables		1	
Coil cable		1	
CAT 6		1	
myRIO PCB		1	
Hex driver		2	
M4 screws		lots	
HM washers	Half moon or D shaped washers	15	One extra
Threaded inserts		20	Three extra

8. Electrical hardware provided

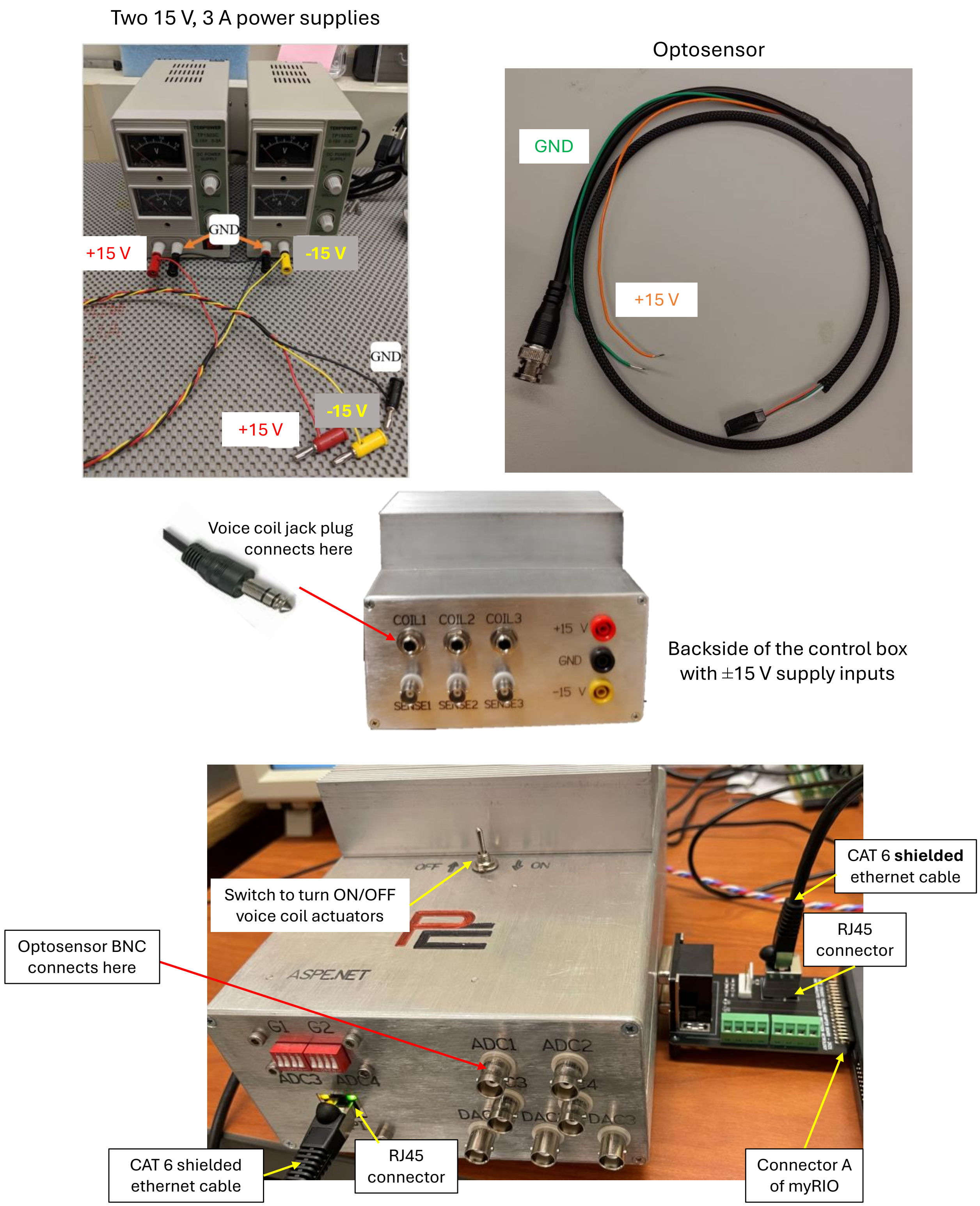
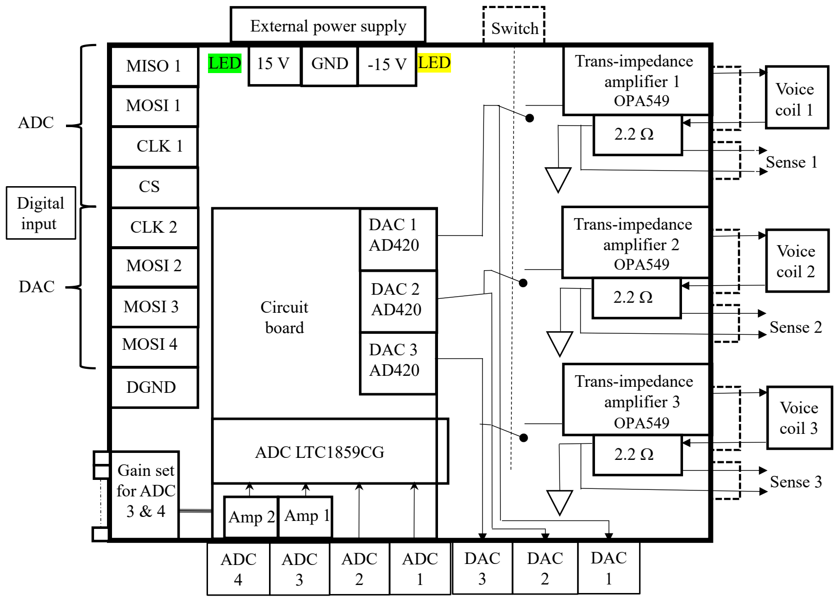


Figure 5: Electrical hardware provided



ADC inputs 1,2,3,4 are BNC connectors
Sense outputs are BNC connectors
Gain setting uses a swithch array.
Digital input is through a Cat6 cable.
Power supply connects using banana plugs.
Jack plugs are used to connect to the voice coils
Switch 'SW' turns off current to all coils.
Power supplies and connecting cables are supplied to the teams.

Figure 6: Block diagram of the 16-bit DAQ and voice coil amplifiers inside the Control box

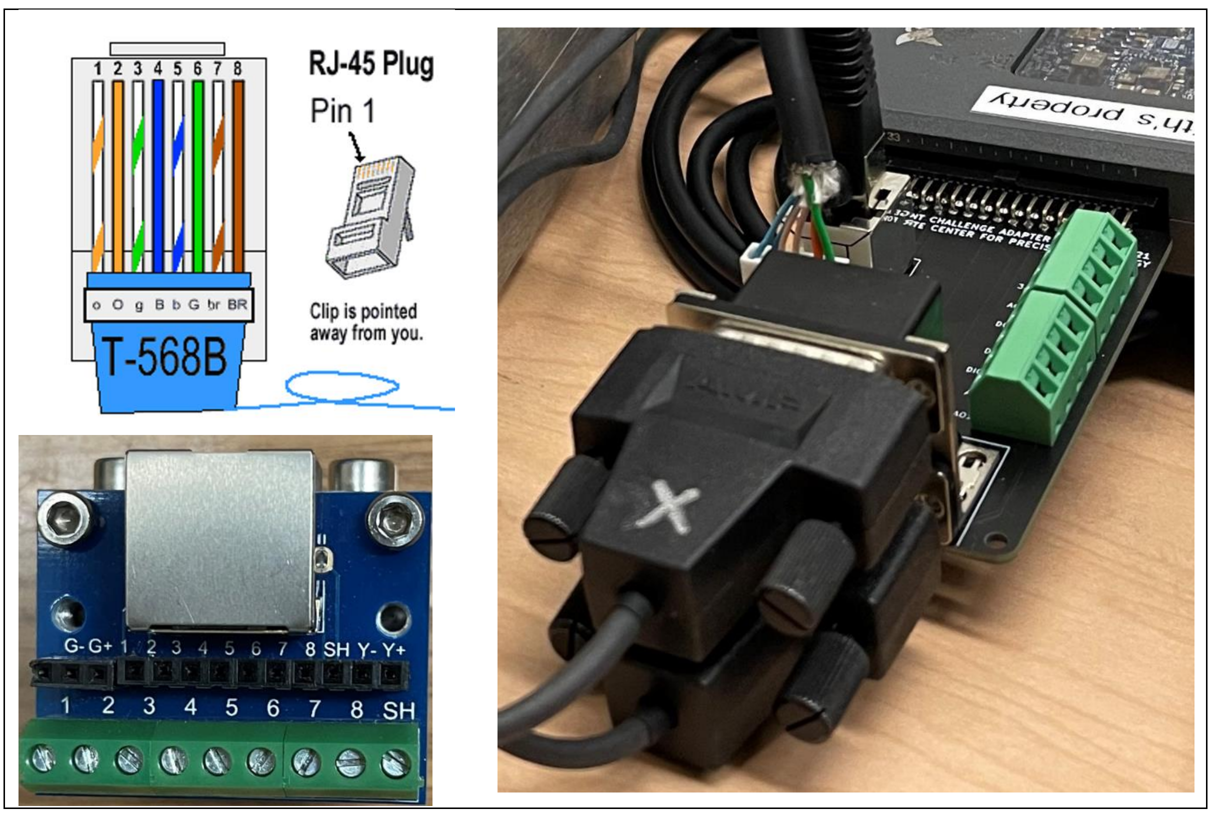
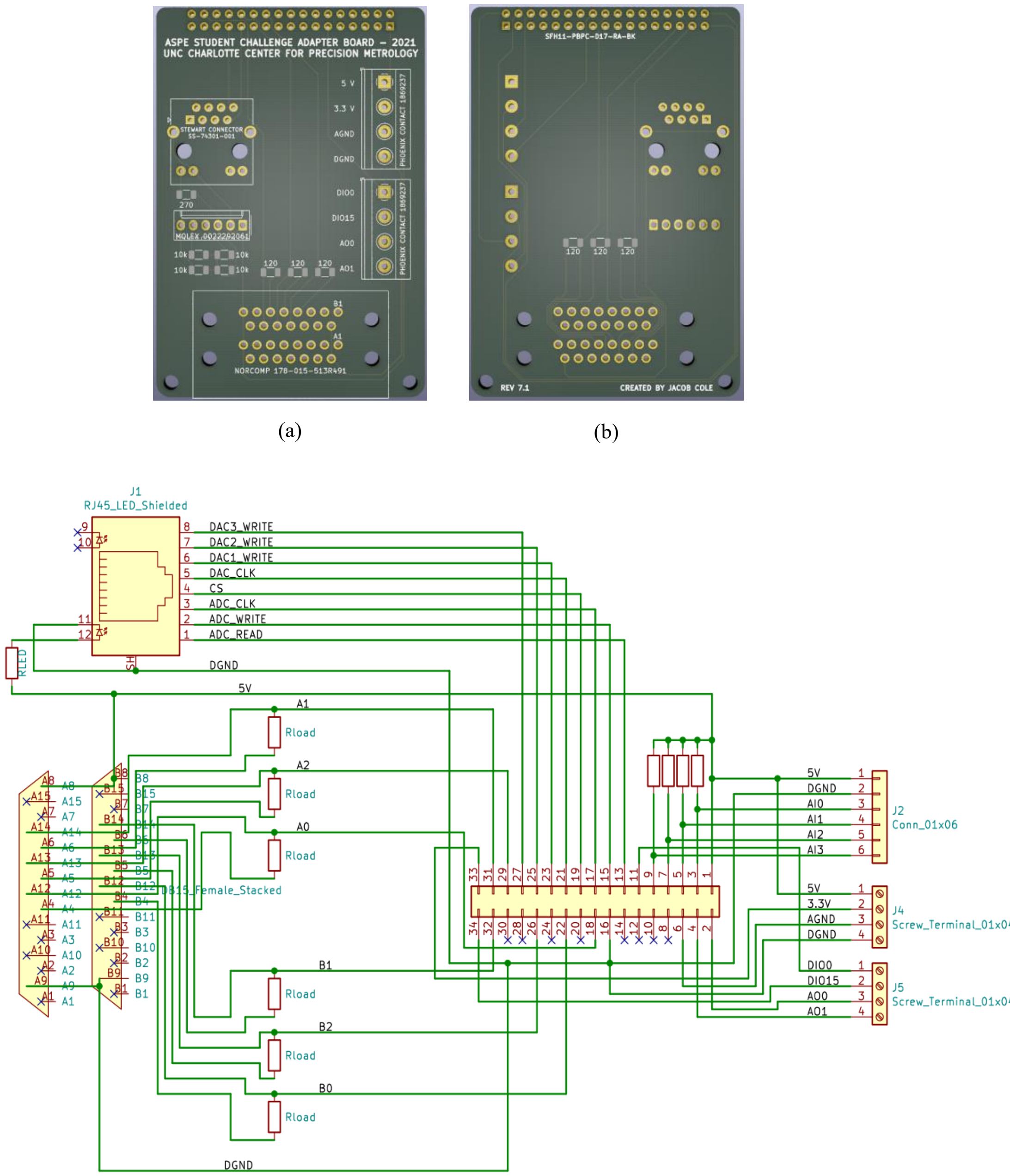


Figure 7: CAT 6 cable connects between one RJ45 breakout board (shown on bottom left) on the front side of the Control box, and the other RJ45 on the PCB (shown on the right side) to interface myRIO and the control box. DB15 connectors are not used for this year's challenge.



(a) (b)

(c)

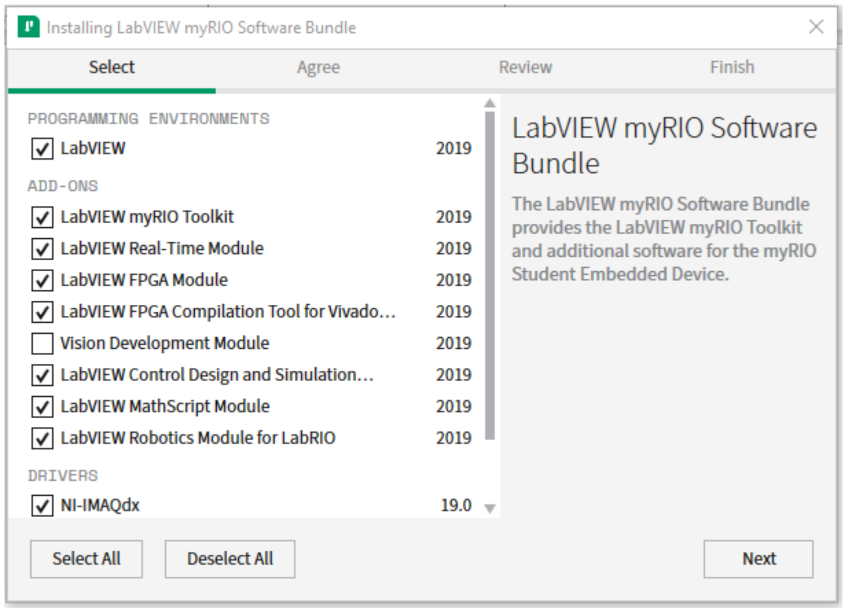
Figure 8: PCB interfacing the Control box and myRIO

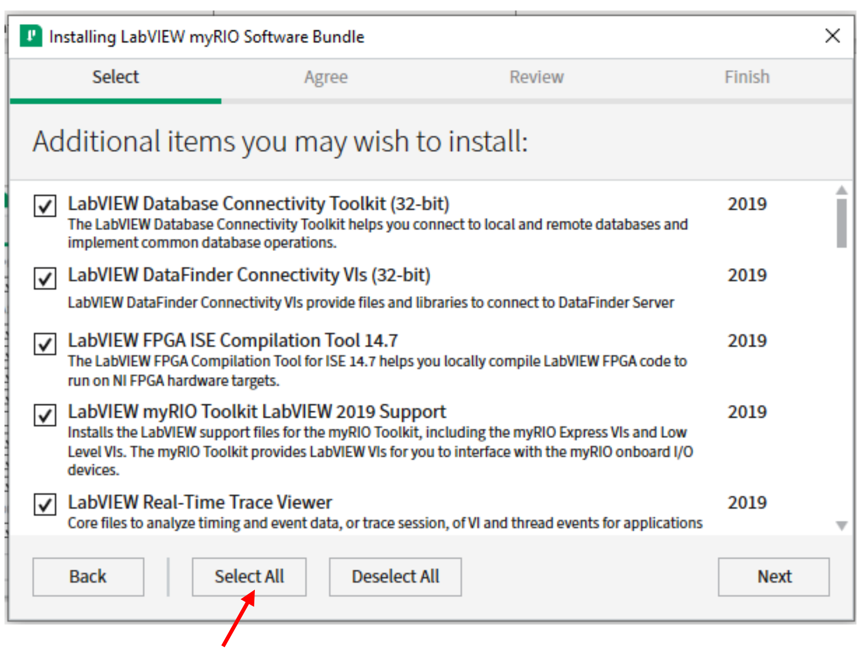
9. LabVIEW installation

(courtesy: Dr. Lei Zhou)

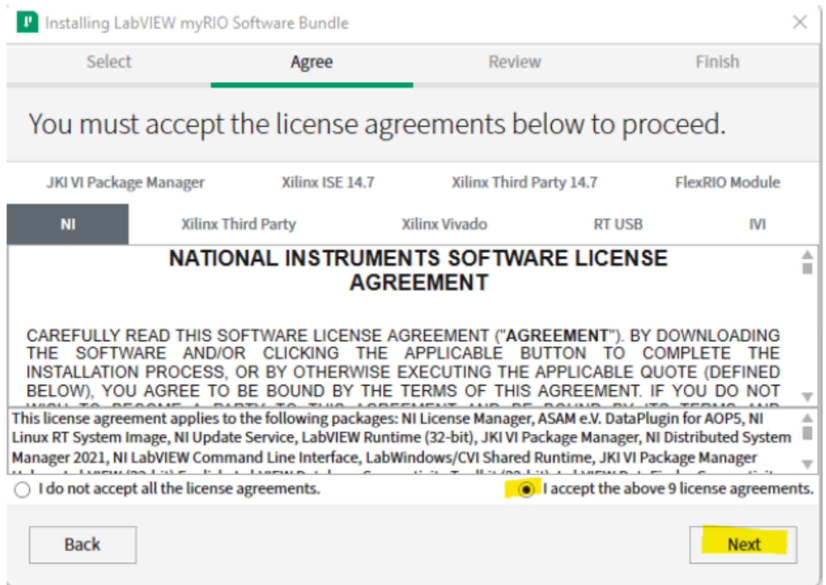
Please make sure to download and use the **LabVIEW 2019** myRIO software bundle from here, which includes the myRIO toolkit, realtime module, and FPGA module necessary for the Student Challenge purposes.

During installation select all exept for Vision development module, as shown below.

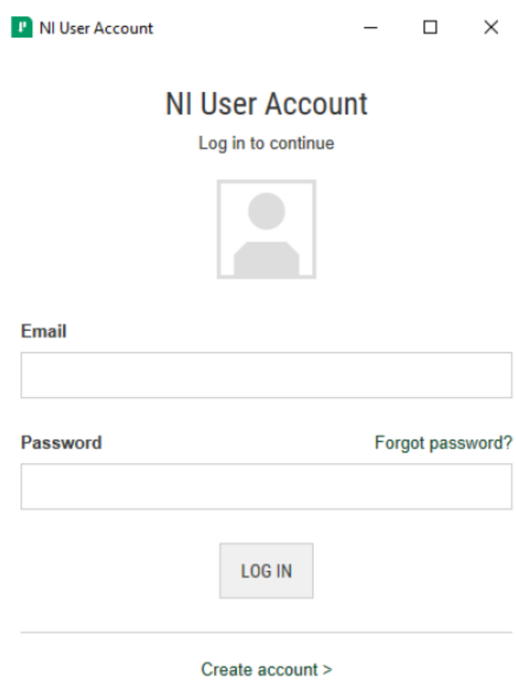




- You will need to accept multiple license agreements for each module.



- The installation process may take one hour. Your computer may need a restart at the end of the installation.
- Once the installation is close to finish, it pops licensing wizard as the screenshot below:

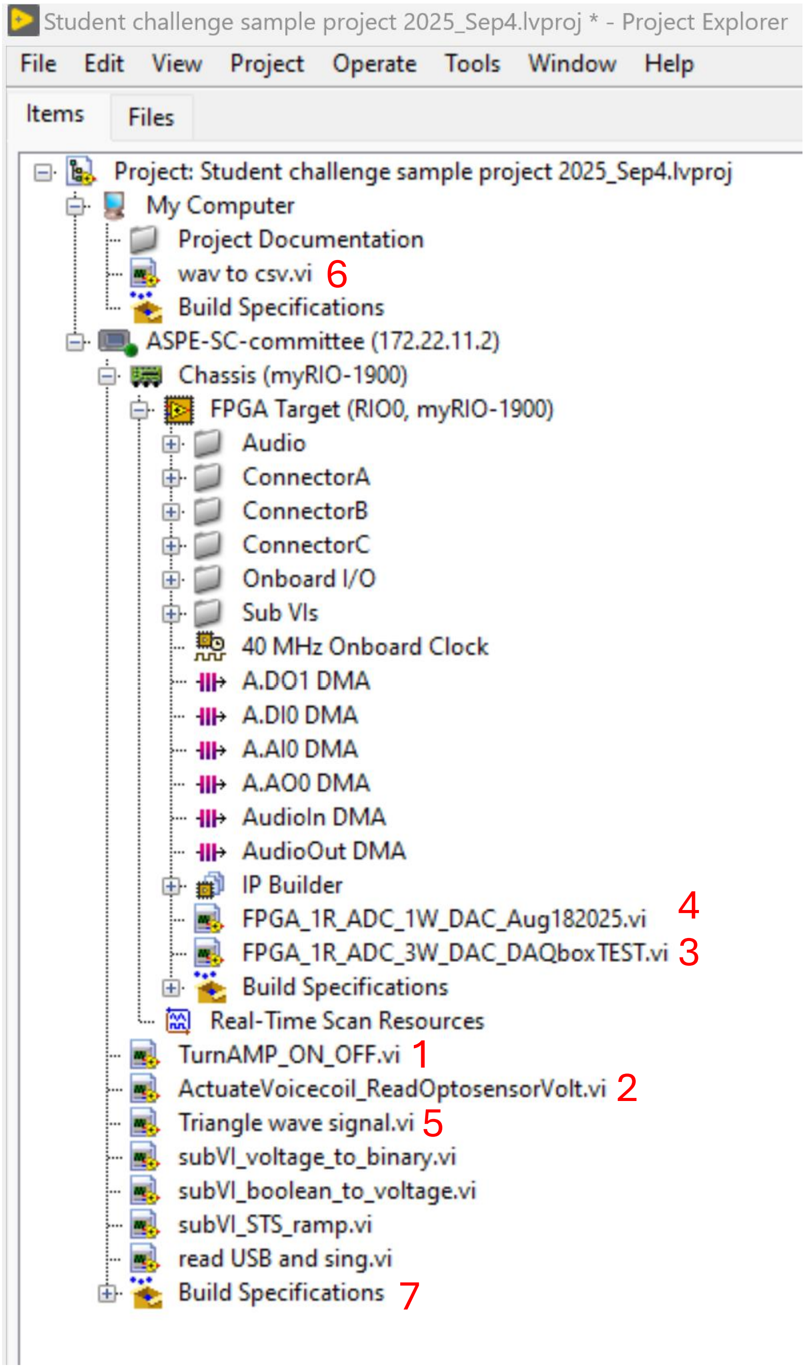


If your school does not have a serial number, LabVIEW lets you evaluate for 7 days, and later for 45 days. Please let us know if you have issues.

If your myRIO has a different firmware version, follow the instructions here.

10. LabVIEW project file

The project file can be found in Github which contains the .vi's that interact with the hardware provided via NI-myRIO, see image below.



- Before turning on (off) the power supply and flicking the switch (on/off) on top of the control box, use program (1) to send 0 V input to the DACs connected to the amplifiers in the control box

To turn on the amplifier:

1. Make sure the control box flick switch and power supply are off
2. Run the program to send 0 V
3. Turn on the power supply
4. Turn on the flick switch
5. You can stop the program

To turn off the amplifier:

1. Run the program to send 0 V
2. Turn off the control box flick switch
3. Turn off the power supply
4. You can stop the program

- Program (2) sends a voltage input to amplifier channel 1 to actuate the voice coil, and read the optosensor voltage.
- Program (5) generates a triangular wave.
- Integrate (2) and (5) to your feedback control algorithm to track the triangular waveform.

Program (6) converts .wav file to .csv file

Program (7) reads the .csv file from the USB and sends the song signal to the voice coil. Integrate this program with your overall control architecture if you prefer.